Answer Key

EECS 489 Midterm Exam

Fall 2024

Name: ______________________________

Uniqname: ______________________________

Instructions

Welcome to the EECS 489 Fall 2024 Midterm.

- This exam is 120 minutes long and consists of both multiple choice and free response questions. Partial credit will be awarded where applicable, so please show all work and reasoning in your responses.
- This is a closed-book exam. You may use one note sheet, 8.5″×11″, double-sided, with your name on it.
- Record all solutions in the space provided. Any answers given outside the boxes will not be graded.
- Clearly fill your multiple choice answers. Erase completely if you change your answer.
- Bubble in your answer on the bar **above** the question. If your circled and bubbled answers are different, we will choose the answer that is bubbled in.
- There are questions on both sides of each page.
- The total points on the exam add up to 80. Your final grade will be out of a 100. You can potentially get 80 out of a 100

Good luck!

When you are finished with the exam, sign below acknowledging the Honor Code

"I have neither given nor received unauthorized aid on this examination, nor have I concealed any violations of the Honor Code."

____

Answer Key

This Page Is Intentionally Left Blank

Multiple Choice

(2 points each)

Ⓐ Ⓑ ● Ⓓ

1. Which of the following strategies does a Content Delivery Network (CDN) utilize to improve the application performance?

I. Optimizing connections using the three “P”s (Persistent, Parallel, Pipelined)
II. Caching through forward and reverse proxies
III. Data replication across multiple servers

a) I
b) I and III
c) II and III
d) III only

Ⓐ ● Ⓒ Ⓓ

2. Which type of DNS record is primarily utilized by Content Delivery Networks (CDNs) to direct traffic to their servers?

a) A Record
b) CNAME Record
c) MX Record
d) NS Record

Explanation: A CNAME record aliases a domain name to another. CDNs use this to transform URLs containing from the client site (e.g. *static01.nyt.com*) to a URL hosted by the CDN (e.g. *nytimes.map.fastly.net*).

Answer Key

Ⓐ Ⓑ ● Ⓓ

3. Which of the following is **not** an advantage of the DNS system?

a) Hierarchical naming structure prevents naming collisions.
b) A layer of indirection above IP addresses enables load-balancing methods.
c) The existence of root servers allow DNS to provide persistent consistency.
d) Hierarchical administration enables the web to scale more easily.

Explanation: The existence of root servers does not enable persistent consistency; in fact, the distributed nature of DNS causes inconsistency, and relying on a few critical root servers creates a limited number of points of failure. The other three options represent real advantages of DNS.

Ⓐ ● Ⓒ Ⓓ

4. Which of the following reasons explains why HTTPs requests over non-persistent connections are not suitable for streaming video media?

a) High bandwidth requirements
b) TCP handshake overhead leading to latency
c) Inability to support secure connections
d) HTTP only supports text-based content

Ⓐ ● Ⓒ Ⓓ

5. What is Dynamic Adaptive Streaming over HTTP (DASH) primarily designed to achieve?

a) Provide secure data transmission over the internet
b) Allow high-quality streaming of media by adapting to network conditions
c) Enable peer-to-peer file sharing without a central server
d) Optimize network routing for data packets

Ⓐ Ⓑ ● Ⓓ

6. In which situation does Dynamic Adaptive Streaming over HTTP (DASH) primarily lower the bitrate of video streaming?

a) Spikes in latency
b) Improved network security
c) Decreased throughput
d) Higher device processing power

Ⓐ ● Ⓒ Ⓓ

7. What term is used to describe the traffic that occurs between servers within a data center?

a) North-South traffic
b) East-West traffic
c) Up-Down traffic
d) In-Out traffic

Ⓐ ● Ⓒ Ⓓ

8. Why is I/O multiplexing (e.g. `select`) often preferred over multi-threading for I/O-heavy tasks in network servers such as the proxy in Project 2?

I. It allows using multiple CPU cores to decrease I/O latency
II. It reduces resource consumption and minimizes context switching.
III. It simplifies error handling in multi-threaded environments.
IV. It guarantees lower latency for all types of tasks.

a) I and IV
b) Only II
c) II and III
d) I, II, III, and IV

Clarification: This question was ambiguously worded, so we accepted both (b) and (c).

● Ⓑ Ⓒ Ⓓ

9. What is the key feature of HTTP Multiplexing that eliminates the head-of-line (HOL) blocking problem?

a) Allowing servers to respond to HTTP GET requests out-of-order.
b) Allowing clients to request multiple objects in a single GET request.
c) Allowing a client to have multiple concurrent HTTP connections with a server.
d) Allowing servers to return partial objects.

Explanation: The HOL blocking problem is created when a large request takes a lot of time to fulfill, holding up the much smaller requests behind it. The solution that eliminates this problem is allowing servers to respond to requests out-of-order (so smaller requests can be dealt with first, for instance). Allowing clients to request multiple objects does not solve this problem on its own, nor does having multiple concurrent HTTP connections (as each connection still faces individual HOL blocking). Returning partial objects, while helpful, is useless for eliminating HOL blocking unless servers can also reorder requests.

Ⓐ Ⓑ ● Ⓓ

10. How many RTTs does it take for HTTP to get the response for a request of a very small file (negligible transmission delay) with no prior TCP connection and **ignoring** the TCP `FIN/ACK` cycle?

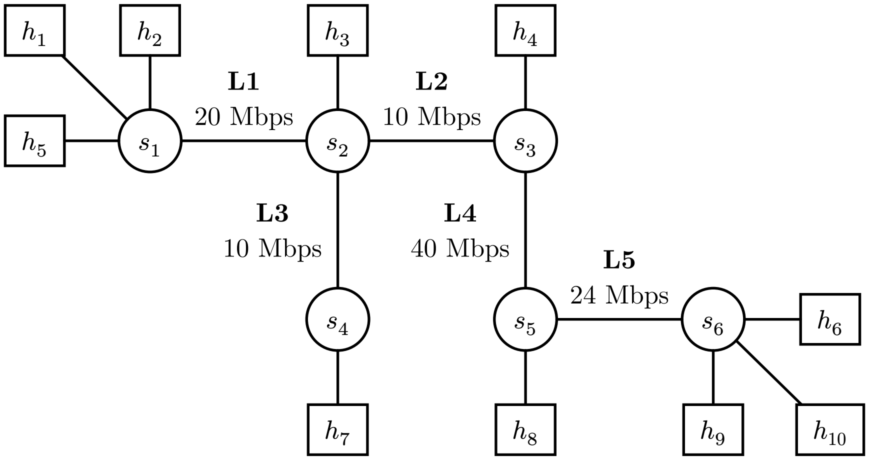
a) 1
b) 1.5
c) 2
d) 2.5

Ⓐ Ⓑ ● Ⓓ

11. HTTP is stateless protocol. Which of the following can help to maintain states?

a) HTTPS
b) DNS cache
c) Cookies
d) Server-side rendering

The next two questions refer to the network topology provided below. You may assume that links between hosts and switches (e.g. $h_1 \longleftrightarrow s_1$) have unlimited bandwidth and zero latency.



Ⓐ Ⓑ Ⓒ ●

12. Consider the given network topology diagram. Suppose that h_1 is sending a large payload to h_{10}.

Suppose in Scenario A, the capacity of intermediate routers ensures that there are no queuing delays. In contrast, in Scenario B, there are queuing delays due to the switches having lower capacity.

Which of the following **from the perspective of h_1** will change between Scenario A and Scenario B?

I. Bandwidth of the connection
II. Throughput of the connection
III. Latency of the connection

a) None of the above
b) III only
c) I and II only
d) II and III only
e) I, II, and III

Answer Key

	Ⓐ Ⓑ Ⓒ Ⓓ

13. Consider the same network topology. Suppose h_1 sends data to h_6, h_2 sends to h_9, and h_4 sends to h_{10} (all simultaneously). **Bandwidth is distributed evenly, and all three hosts are sending large payloads.**

Which of the answers is closest to the maximum throughput h_1 can observe when sending the payload?

a) 3.3 Mbps
b) 12 Mbps
c) 10 Mbps
d) 15 Mbps
e) 5 Mbps

Short Answer

14. Bandwidth and Latency

Efe is requesting a 960MB file from a server. Suppose 5.0 seconds pass between the server beginning to send the file and Efe receiving the full file. Efe then requests a 480MB file from the same server; 2.6 seconds pass in this case (between the server beginning to send the file and Efe receiving the full file).

Assume there are no queuing or processing delays on this network.

What is the estimated **bandwidth and RTT** of the network connection between Efe's computer and the server? Please justify your answer and show any intermediate calculations.

Answer: As there are no queuing or processing delays on this network, the amount of time it takes to send a file is determined entirely by the propagation delay and the transmission delay. Observe that the propagation delay is constant with respect to the size of the packet, while the transmission delay scales linearly with the size of the packet.

Let b be the bandwidth of the link and l be the propagation delay (latency). Then,

$$\frac{480 \text{ MB}}{b} + l = 2.6s$$

and

$$\frac{960 \text{ MB}}{b} + l = 5.0s.$$

Solving for b and l, we have $b = 200$ MB/s and $l = 0.2s$. Then, we can estimate the bandwidth as **1600 Megabits per second (Mbps)** and the latency as **400 ms**.

Answer Key

This Page Is Intentionally Left Blank

15. TCP Sequence Numbers

Suppose a series of packets are being sent over a TCP connection from Host A to Host B. Fill in the table below with the appropriate sequence numbers. For the last column, please either specify the sequence number of the ACK, or state that no ACK was sent.

Packet Number	Packet Sent (Time)	Packet Received (Time)	Data Size (Bytes)	Packet Sequence Number	ACK Sequence Number
1	0	1	20	40	60
2	1	2	10	60	70
3	2	Not Received	30	70	No ACK Sent
4	3	4	40	100	70
3 (Resend)	7	8	30	70	140

16. TCP Congestion Control

(24 points)

We now consider the congestion control mechanisms of a TCP connection. Suppose CWND begins at 1 packet and ssthresh is initially set to 16 packets.

When a timeout occurs, the following updates occur:
1. ssthresh = CWND / 2
2. CWND = 1 packet

On a successful ACK, the following updates occur:

1. CWND += 1 if CWND < ssthresh
2. CWND += $\frac{1}{\lfloor \text{CWND} \rfloor}$ if CWND $\geq$ ssthresh

We do not consider duplicate ACKs for the purposes of this question. Assume that the latency of the network is nearly constant; as such, we measure time in units of RTT, starting with 0 RTT. Assume packets are small enough in size to be sent out nearly instantaneously.

16.1 *(8 points)* Assume that the network never experiences any packet loss or packet reordering. Please fill out the following table for CWND over time. For every timestamp, assume that all ACKs from the previous round of packets have been received, but no new packets have been sent out yet.

Time (in RTT from start)	**0**	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**
CWND (in packets)	1	2	4	8	16	17	18	19	20

16.2 *(6 points)* Now, consider the scenario presented in (16.1), except the receiver window size (RWND) is set to 8 packets. Please fill out the table for how CWND increases over time under this scenario. Note that CWND can be fractional.

Time (in RTT from start)	**0**	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**
CWND (in packets)	1	2	4	8	16	16.5	17	$17\frac{8}{17}$	$17\frac{16}{17}$

Common Error: Students sometimes incorrectly assumed that RWND is not known to the sender (in fact, it is communicated as a part of TCP). This caused them to infer that a packet drop occurs once CWND becomes larger than 8.

Common Error: Students sometimes incorrectly assumed that CWND is upper-bound by RWND.

16.3 *(5 points)* What will be the measured throughput by the sender in the scenario described in the above question from $t = 0$ RTT to $t = 5$ RTT? Your units should be $\frac{\text{packets}}{\text{RTTs}}$.

Clarification: The intention of this question was to be over the first five rounds of packets. Due to some ambiguity in the question, full credit was awarded if a student calculated throughput over the first six rounds of packets instead.

Answer: Notice that the actual number of packets sent will be $\min(\text{RWND}, \text{CWND})$. So, the actual number of packets sent is $1 + 2 + 4 + 8 + 8 = 23$. As this occurs over 5 RTTs, the throughput is measured as $\frac{23}{5} = 4.6 \frac{\text{packets}}{\text{RTT}}$.

Common Error: Students sometimes calculated the total number of packets over six rounds and then divided by five (or calculated the total number of packets over five rounds and divided by six). This generally earned -1 point, as they needed to be consistent with their interpretation of the question.

Common Error: Students sometimes interpreted CWND as the actual number of packets sent. The actual number of packets sent is min(CWND, RWND).

16.4 *(5 points)* Now, consider the scenario presented in (16.1), except assume that there is a sender-side timeout triggered right before $t = 4$ RTT for the packets sent out at $t = 3$ RTT, which are dropped by the network. Fill out the following table for CWND over time.

Time (in RTT from start)	**0**	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**
CWND (in packets)	1	2	4	8	1	2	4	5	6

Common Error: Not setting ssthresh to (CWND/2 = 4) after the packet drop.

17. DNS Takeover

(14 points)

Joyce tries to access her bank's login page at `login.bank.com` to make a deposit while at her office. Unfortunately for Joyce, her bank does not use HTTPS, meaning that her browser cannot verify the identity of the server that he connects to.

Joyce's office's network components include: a router, a switch, and a DNS server.

You may assume that no one at Joyce's office has recently accessed any `.com` domains

17.1 *(10 points)* Describe, numbered in order, the requests and responses for the DNS lookup that will occur when Joyce tries to access `login.bank.com`. You should assume that the DNS queries are done recursively.

You should consider that the DNS server for `bank.com` is authoritative for all domains in the form `*.bank.com`.

If any of the requests/responses are from/to the root server, simply list "Root server" as the destination/source. The same applies for requests/responses from/to the Local DNS.

Fill in the table below. You may not need to use all of the rows.

#	Source	Destination	Request/ Response (Req/Res)
1	Local DNS	Root Server	Req
2	Root Server	.com	Req
3	.com	bank.com	Req
4	bank.com	.com	Res
5	.com	Root Server	Res
6	Root Server	Local DNS	Res

17.2 *(4 points)* To take advantage of this situation, Mallory has set up her computer between Joyce's office's DNS server and the rest of the internet. Her setup allows her to listen to and send packets from/to the DNS server on Joyce's network. However, Mallory does not have direct access to the link between Joyce's computer and her office's local network components.

Describe one way in which Mallory use these facts to steal Joyce's banking information.

Answer: When Joyce's computer sends a DNS request for login.bank.com, Mallory sees the request. Before the actual DNS server can respond, she can respond with a fake IP address that points to her malicious server, which is designed to look exactly like the bank's login page. When Joyce visits the page, thinking it's legitimate, she enters her banking credentials, which Mallory collects.

18. Load Balancing Nameserver

The nameserver you implemented for Project 2 used basic load balancing methods such as Round-Robin. Now, we would like to implement more advanced load balancing on our nameserver.

Each of the content servers now listen on a socket for a "load check" command, to which they respond with how much load they currently are under. The "load check" command is simply a single-byte transmission of the character `L`.

The servers respond by sending a 4-byte, network byte-order integer 0-100 that represents how much load the server is under (100 being the maximum).

Below, implement the function `select_lowest_load_server`. The function should

- Query all content-server sockets in the `sockets` array for their load levels
- Return the server socket with the minimum load-level
- The queries need not be done in parallel

The following function signatures may be helpful:

```
ssize_t send(int sockfd, const void buf[.len], size_t len, int flags);
ssize_t recv(int sockfd, void buf[.len], size_t len, int flags);
```

`MSG_WAITALL` is defined, and you may use it if you desire. You can also assume that `send` is guaranteed to transmit `len` number of bytes on a single call.

Answer:

```
int select_lowest_load_server(const std::vector<int>& sockets)
  uint32_t min_load = 101, size_t min_ind;
  for (size_t i = 0; i < sockets.size(); ++i) {
    send(sockets[i], "L", 1, 0);

    uint32_t server_load;
    recv(sockets[i], &server_load, sizeof(server_load), MSG_WAITALL);
    server_load = ntohl(server_load);

    if (server_load < min_load) {
      min_load = server_load;
      min_ind = i;
    }
  }

  return sockets[min_ind];
```

Answer Key

19. Video Playback

(16 points)

You want to watch a video. The network between your computer and the video server has a bandwidth of **80 Megabits per Second (Mbps)** and a latency of **500ms**. The video is **5 minutes** long with **30 frames per second (fps)**, with each frame taking a constant **500 Kilobytes (KB)** to store. Assume that your computer requires a buffer of **5 seconds** of video in order to begin playout, following a standard model for video streaming as covered in lecture.

Assume:

- The software for sending video frames on the server-side and receiving video frames on the client-side are both fast enough that the network is the bottleneck.
- Once the server begins sending the video, it continues to push video data into the network as fast as the network will allow until the whole video is sent.

As always, justify your answers and show any intermediate calculations for the following questions.

Clarification: We applied carryover error grading rules for all parts of this question. In other words, a student's answer for a particular part earned full credit if it was consistent with their answers to previous parts.

19.1 *(4 points)* How many megabytes of video need to be buffered to begin playback?

Answer: We need at least

$$\left(5s \cdot 30\frac{\text{frames}}{s} \cdot 0.5\frac{\text{MB}}{\text{frame}}\right) = 75 \text{ MB}$$

of video buffered for playback to begin.

19.2 *(3 points)* Assume the video server begins sending data at time $t_0 = 0$. At what time t_1 does the first bit of data arrive at your computer (in seconds)?

Answer: The propagation delay between the video server and your computer is 500 ms, so $t_1 - t_0 = 0.500s$.

Common Error: Students misinterpreted the latency being 0.500s as the RTT being 0.500s.

19.3 *(3 points)* The video begins playback at time t_2. Calculate $t_2 - t_1$ (in seconds).

Answer: The rate that we can receive frames at is

$$\frac{80 \text{ Mbps}}{8 \text{ bits/byte}} = 10 \text{ MBps}.$$

Thus, it takes

$$\frac{75 \text{ MB}}{10 \text{ MBps}} = 7.5 \text{ seconds}$$

to get 75 MB worth of video in the buffer, assuming it starts out completely empty.

19.4 *(3 points)* Playback is interrupted as the buffer is empty for the first time at time t_3. Calculate $t_3 - t_2$ (in seconds).

Answer: First, notice that we are playing back video at 30 FPS; as each frame takes 0.5 MB of space, the rate of video playback is

$$30 \frac{\text{frame}}{\text{second}} \cdot 0.5 \frac{\text{MB}}{\text{frame}} = 15 \text{ MBps}.$$

Once the video begins playing, the amount of time it takes for the buffer to empty again will be

$$\frac{75 \text{ MB}}{15 \text{ MBps} - 10 \text{ MBps}} = 15 \text{ seconds}.$$

19.5 *(3 points)* The video finishes playing on your computer at time t_4. Calculate $t_4 - t_0$.

Answer: The buffer will alternate between filling up to the threshold of 75 MB and draining back to empty as the video is played out, at which point the cycle repeats. Based on the answers to the previous two parts, every iteration of this cycle takes $7.5s + 15s = 22.5s$. Thus, every $15s$ of video takes $22.5s$ to play out. As the video has total duration 300 seconds, it will take

$$300 \text{ seconds} \cdot \frac{22.5 \text{ seconds}}{15 \text{ seconds}} = 450 \text{ seconds}$$

to play back $(t_4 - t_1)$. Finally, we must account for the initial propagation delay $(t_1 - t_0)$ caused by the link, which is a one-time cost. Thus, the actual total amount of time will be

$$450.5 \text{ seconds}.$$

Common Error: Students forgot to include the initial latency in their calculation (extra 0.5 seconds).

Common Error: Students correctly figured out the cyclic behavior of the buffer, but incorporated the latency into every cycle. In other words, every cycle was assumed to be 22.5 + 0.5 = 23s. This is incorrect; as the video server does not stop sending data in the middle of the video, the latency is a one-time cost.

Answer Key

This Page Is Intentionally Left Blank

Answer Key

This Page Is Intentionally Left Blank